



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2024-2025
Digital Learning Handout

Part A: Content Design

Course Title	Fair, Accountable, Transparent Machine Learning
Course No(s)	AIMLCZG517
Credit Units	4
Credit Model	3-1-0
Course Author	Sugata Ghosal
Version No:	2.0
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Course Description:

Biases and fairness, fair representation learning, Interpretability and Transparency, Example and Visualization Based Methods for Interpretability, Interpreting deep neural networks, Fairness Through Input Manipulation, Fair NLP/Vision, Robustness and adversarial attacks/defence, ML auditing, privacy

Course Objectives

No	Course Objective
CO1	Introduce students to the concepts of bias and fairness and techniques for incorporating these in ML
CO2	Introduce students to the concepts of interpretability and transparency and techniques for incorporating these in ML
CO3	Introduce students to the concepts of robustness and techniques for robust ML
CO4	Introduce students to the concepts of privacy in ML

Text Book(s):

T1	Barocas, Solon, Moritz Hardt, and Arvind Narayanan. Fairness and Machine Learning , 2018.
T2	Molnar, Christoph. Interpretable machine learning , 2019.

Reference Book(s) & other resources:

R1	Christopher M. Bishop, Pattern Recognition & Machine Learning, Springer, 2006
R2	https://www.borealisai.com/research-blogs/tutotial*

Learning Outcomes: Students will be able to

	Learning Outcomes
LO1	Gain a strong understanding of the foundations of fairness, interpretability and transparency in Machine Learning algorithms





LO2	Mitigate fairness issues in high-stakes ML applications
LO3	Develop explainability techniques for ML applications
LO4	Mitigate security and privacy issues in ML applications

Modular Content Structure

1. Introduction

- Review - traditional performance criteria for ML
- Emerging requirements
 - Fairness
 - Accountability
 - Transparency
 - Privacy
 - Robustness
- Motivating use cases

2. Fairness and Bias

- Sources of Bias
- Real world examples
 - school admissions, criminal justice, hiring, gender/occupation bias
- Sensitive Features
- Fairness through unawareness

3. Learning Fair Representations

- Major Fairness criteria
 - Direct Solution Method
 - Demographic Parity
 - Equality of Odds/Opportunity
 - FICO Case Study
- Prejudice Removing Regularizer
 - Prejudice index (PI)
 - Optimizing PI
- Adult Income Case Study

4. Fairness thru input manipulation

- Basic Data Manipulation Techniques
 - Reweighting
 - Universal Sampling
 - Preferential Sampling
- Individual Fairness
- Optimized Pre-processing
- Learning to Defer

5. Fair Casual Reasoning

- Causal Fairness and Inherent Bias
- Counterfactual Fairness
 - Formal Methods
 - Case Study – Success in Law School
 - Case Study - Crime Rates and Arrest





- Equalized Counterfactual Odds
- Multiple Causal Worlds
- 6. Fairness in NLP, Computer Vision**
 - Biases in NLP Models
 - Data Augmentation
 - Debiasing Word Embedding
 - Counterfactual Fairness
 - Counterfactual Face Attribution
 - Gender Equalized Image Captioning
 - Adversarial Removal of Gender Features
- 7. Interpretability and Transparency**
 - ML Interpretability
 - Intrinsically Interpretable Models
 - Simple interpretable models
 - Intrinsically interpretable techniques for deep learning
 - Interpretability Concepts
 - Intrinsic and post hoc methods
 - model-specific and model-agnostic methods
 - Local and global interpretable methods
 - Interpretability and performance trade-offs Instance-based Learning
- 8. Feature interaction for interpretability**
 - Feature Interaction
 - Layerwise Relevance Propagation
 - DeepLift
 - Shapley Additive Explanations (SHAP)
 - Coalitional Game and Shapley Values
 - Kernel SHAP
 - Deep SHAP
 - Tree SHAP
 - Equitable Value of Data
- 9. Example and Visualization Based Methods for Interpretability**
 - Example Based Methods
 - Counterfactual Explanations
 - Targeted counterfactual examples
 - Untargeted counterfactual examples
 - Contrastive Examples
 - Concept Based Methods
- 10. Interpreting Deep Networks**
 - Visualization Based Methods
 - Activation Visualization
 - Saliency Maps
 - GoogLeNet Activation Atlas
 - Interpretability via Activation Visualization
 - Gradient Based Feature Attribution
 - Integrated Gradient





- Baselines for Integrated Gradient

11. Robustness and Adversarial Attacks / Defense

- Adversarial Attack
- White-box Evasion Attack
 - FGSM
 - C&W
 - Physical Attack
- Transferability of Attack
- Black-box Evasion Attack
 - Jacobian-based Data Augmentation
- Adversarial Defense
- Defense Strategies
 - Adversarial Training
 - Input Transformations
 - Stochastic Gradients
 - Obfuscated Gradients and BPDA
- Robust Optimization
- Certified Defense

12. ML Auditing and Privacy

- ML Auditing
 - Distill-and-Compare
- Privacy in ML
 - Differential Privacy with Deep Learning
 - Model Inversion Attack and Differential Privacy
 - Local Differential Privacy
 - Federated Learning

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	Introduction	Review - traditional performance criteria for ML; Emerging requirements such as Fairness, Accountability, Transparency, Privacy, Robustness, Motivating use cases	Lecture Slides
2	Fairness and Bias	Sources of Bias, Real world examples such as school admissions, criminal justice, hiring, gender / occupation bias, Sensitive Features, Fairness through unawareness	Lecture Slides
3	Learning Fair Representations	Major Fairness criteria such as Demographic Parity, Equality of Odds/Opportunity, FICO Case Study, Prejudice Removing Regularizer, Adult Income Case Study	Lecture Slides





4	Fairness thru input manipulation	Basic Data Manipulation Techniques, such as Reweighting, Universal Sampling, Preferential Sampling, Individual Fairness, Optimized Pre-processing, Learning to Defer	Lecture Slides
5	Fair Casual Reasoning	Causal Fairness and Inherent Bias, Counterfactual Fairness, Case Study – Success in Law School, Crime Rates and Arrest, Equalized Counterfactual Odds, Multiple Causal Worlds	Lecture Slides
6	Fairness in NLP, Computer Vision	Biases in NLP Models. Data Augmentation, Debiasing Word Embedding, Counterfactual Fairness, Counterfactual Face Attribution, Gender Equalized Image Captioning, Adversarial Removal of Gender Features	Lecture Slides
7	Disentangled Fair Representation	Disentangled Representations, Flexibly Fair Representation, Orthogonal Disentangled Fair Representations, Measurements for Disentangled Fair Representations	Lecture Slides
8	Review	Review of Session 1 to 7	Lecture Slides
9	Interpretability and Transparency	ML Interpretability, Intrinsically Interpretable Models, Interpretability Concepts such as Intrinsic and post hoc methods, model-specific and model-agnostic methods, Local and global interpretable methods, Interpretability and performance trade-offs Instance-based Learning	Lecture Slides
10	Feature Interaction for Interpretability	Feature Interaction such as Layerwise Relevance Propagation, DeepLift, Shapley Additive Explanations (SHAP) and variants	Lecture Slides
11	Example Based Methods	Example Based Methods, Counterfactual Explanations, Contrastive Examples, Concept Based Methods	Lecture Slides
12	Interpreting Deep Networks	Visualization Based Methods, Activation Visualization, saliency maps, Interpretability via Activation Visualization, Gradient Based Feature Attribution ○	Lecture Slides
13	Robustness and Adversarial Attacks	Adversarial Attack, White-box Evasion Attack, Transferability of Attack, Black-box Evasion Attack	Lecture Slides
14	Robustness and Adversarial Attacks	Adversarial Defense, Defense Strategies such as Adversarial Training, Input Transformations, Stochastic Gradients, Obfuscated Gradients	Lecture Slides





15	Auditing and Privacy	ML Auditing, Distill-and-Compare, Privacy in ML, Differential Privacy with Deep Learning, Model Inversion Attack and Differential Privacy, Local Differential Privacy, Federated Learning	Lecture Slides
16	Review	Review of session 9 to 15	Lecture Slides

Experiential Learning Components:

Describe objective, outcome of Experiential Learning Component and the lab infrastructure needed (virtual, remote, open source etc...) number of lab exercises needed, etc.

1. Lab work: 6
2. Project work: 0
3. Case Study: 4 Webinars
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on implementation of FAccT techniques using state of art tools

Scope of Experiential Learning Component:

Programming language - Python

Tools and libraries: Jupyter, ScikitLearn, Keras, Tensorflow etc

Lab Infrastructure:

Online/ Open source/Google Colab

List of Experiments:

Lab No	Lab Objective	Session Reference
1	Detecting bias in ML model using different fairness metrics	3
2	Bias mitigation in ML model using disparate impact remover	3
3	Improving the Fairness of a Classifier using Prejudice Removal Regularizer	3
4	Implementing LIME and SHAP algorithms to interpret different machine learning classifiers	9
5	Visualizing and interpreting a CNN based Image classifier	12
6	Implementing Adversarial attacks on CNN based Image classifier	14

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session





Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weight	Duration	Day, Date, Session, Time
EC – 1*	Quiz	Online	10%	1 week	To be announced
	Assignment/Lab Assignment / Lab Exams	Online	20 %	10 days	To be announced
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	To be announced
EC - 3	Comprehensive Exam	Open Book	40%	2 ½ Hours	To be announced

EC1* (20% - 30%): Quiz (optional): 5-10 %, Lab Assignment/Assignment: 20% - 30%

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: <https://elearn.bits-pilani.ac.in>

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: “open book” means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.
[EC3 Guidelines](#)
4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

